



The University of Manchester

BERT: Bidirectional Encoder Representations from Transformers

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Transformer

Encoder Only

Uses: NLU, classification, feature extraction

Examples: BERT, RoBERTa, DeBERTa

Encoder-Decoder

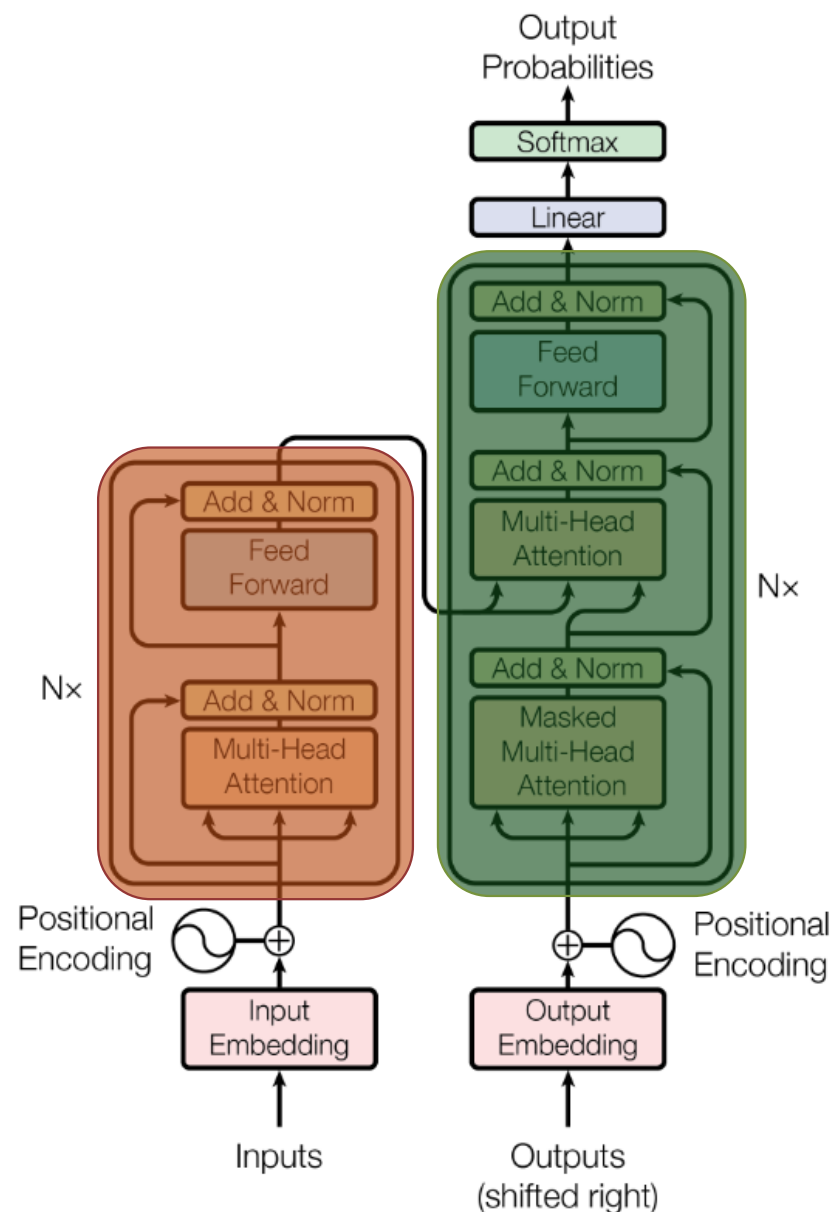
Uses: NLG, translation, summarisation, seq2seq mapping

Examples: T5, Flan-T5, BART

Decoder Only

Uses: NLG, translation, summarisation, completion

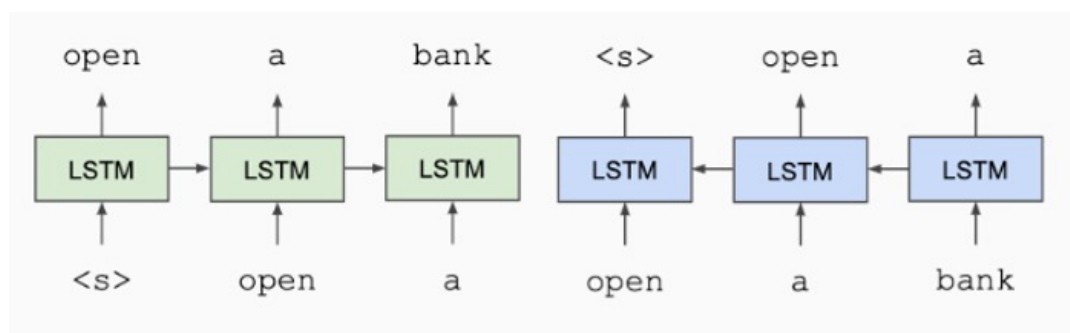
Examples: GPT-x, LLaMa, every other new model today



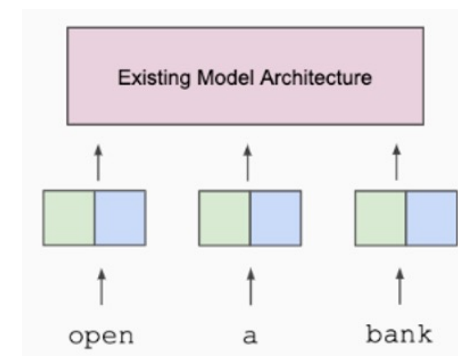
Prior BERT: ELMo LSTM based

ELMo (Peters et al., 2018; NAACL 2018 best paper)

- Train two separate unidirectional LMs (left-to-right and right-to-left) based on LSTMs
- Feature-based approach: pre-trained representations used as input to task-specific models
- Trained on single sentences from 1B word benchmark (Chelba et al., 2014)



Train separate left-to-right and right-to-left LMs



Apply as "Pre-trained Embeddings"

cons: slow to change, not good for downstream task, hard to parallel.

BERT: Key Contributions



- A deep **Transformer encoder** model pre-trained on large corpora
- The key: learning representations based on **bidirectional context**

Why? Because both left and right contexts are important to understand the meaning of words.

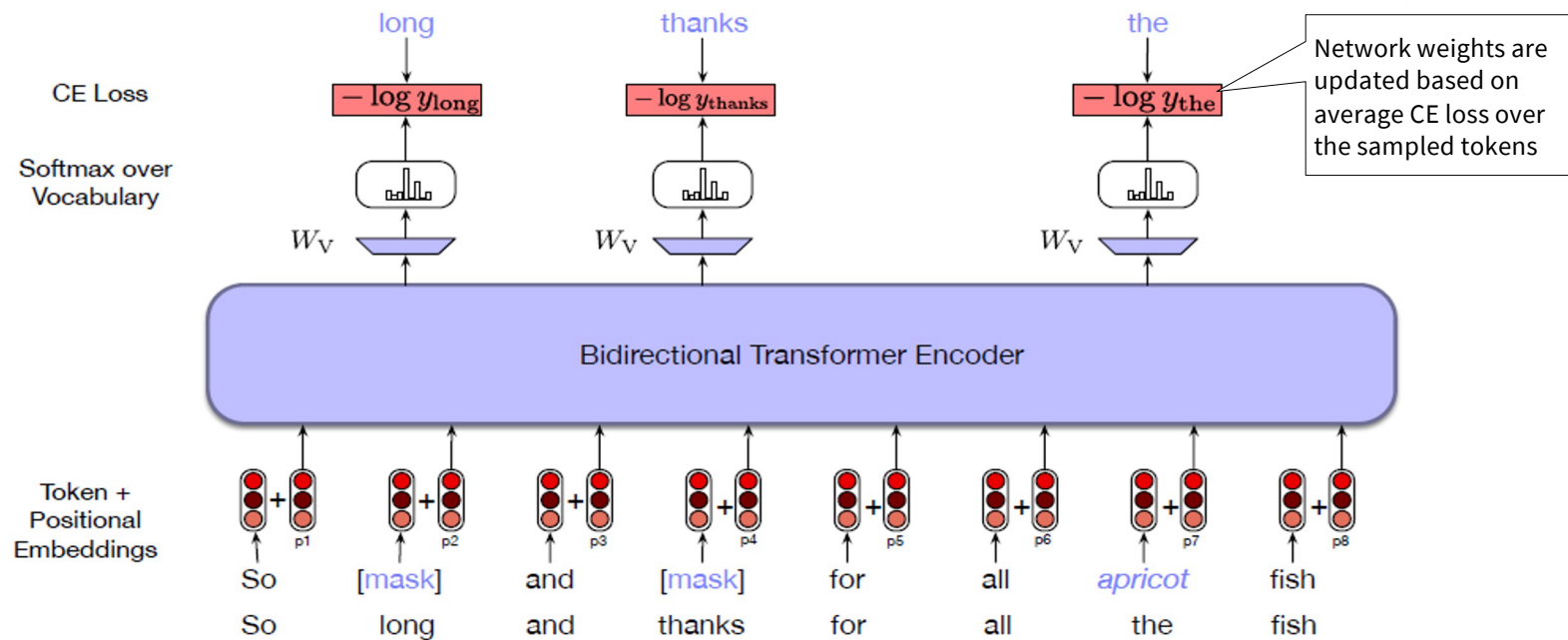
Example #1: we went to the river bank.

Example #2: I need to go to bank to make a deposit.

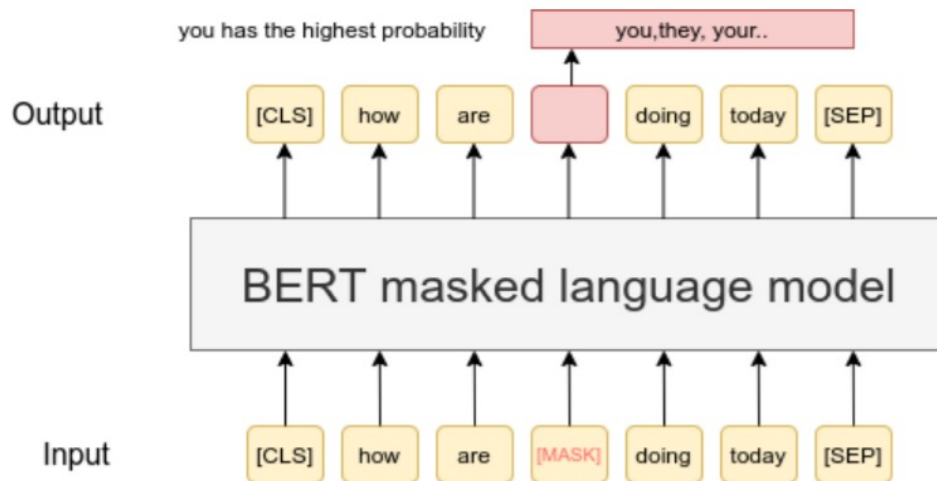
- **Pre-training objectives:** (1) masked language modeling + (2) next sentence prediction,
- **State-of-the-art performance (2018)** on a large set of sentence-level and token-level tasks.

Masked Language Modelling (MLM)

- All input tokens participate in bidirectional self-attention BUT only the 15% of tokens sampled for prediction contribute to the training loss.
- Average cross-entropy loss over these selected positions



Masked Language Modelling



selected mask token contribute to predunction

Examples

- He is from Kuala Lumpur, Malaysia.
→ He [MASK] from Kuala [MASK], Malaysia.
- He went to the store.
→ He went to the [MASK].

MLM: Masking Rate and Strategy

- Q: What is the value of k ?
 - They always use $k = 15\%$.
 - Too little masking: computationally expensive
 - Too much masking: not enough context
- Q: How are masked tokens selected?
 - 15% tokens are uniformly sampled
 - Is it optimal? See span masking (Joshi et al., 2020)
 - Example: He [MASK] from Kuala [MASK], Malaysia.

MLM: 80-10-10 Corruption

- For the 15% of tokens selected for prediction,
 - 80% of the time, they replace it with [MASK] token
went to the store → went to the [MASK]
 - 10% of the time, they replace it with a random word in the vocabulary
went to the store → went to the running
 - 10% of the time, they keep it unchanged
went to the store → went to the store

Why? Because [MASK] tokens are never seen during fine-tuning

Next Sentence Prediction (NSP)

- **Motivation:** many NLP downstream tasks require understanding the relationship between two sentences (e.g. NLI, paraphrase detection, QA)
- **NSP** is designed to reduce the gap between pre-training and fine-tuning

[CLS]: a special token
always at the beginning

[CLS]: a special token used to separate two segments

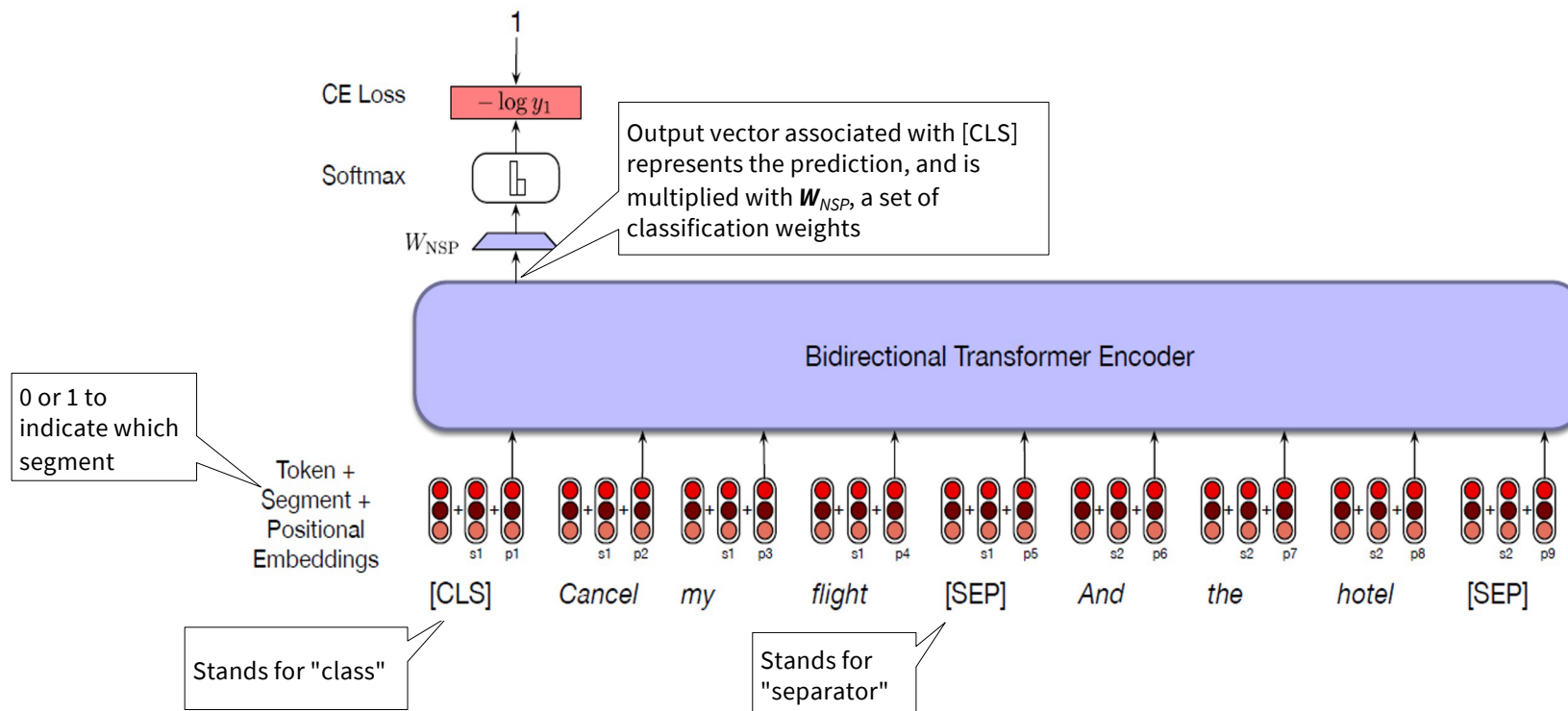
Input = [CLS] the man went to [MASK] store [SEP]
he bought a gallon [MASK] milk [SEP]

Label = IsNext

```
Input = [CLS] the man [MASK] to the store [SEP]
        penguin [MASK] are flight ##less birds [SEP]
```

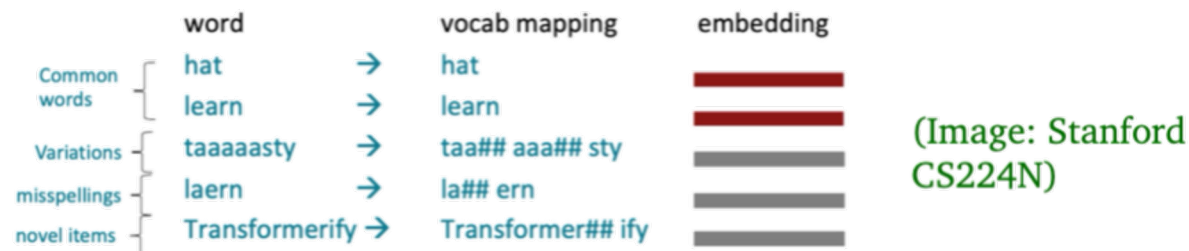
Label = NotNext

Next Sentence Prediction (NSP)



BERT pre-training: Putting together

Vocabulary size: 30,000 word pieces (common sub-word units) ([Wu et al., 2016](#))

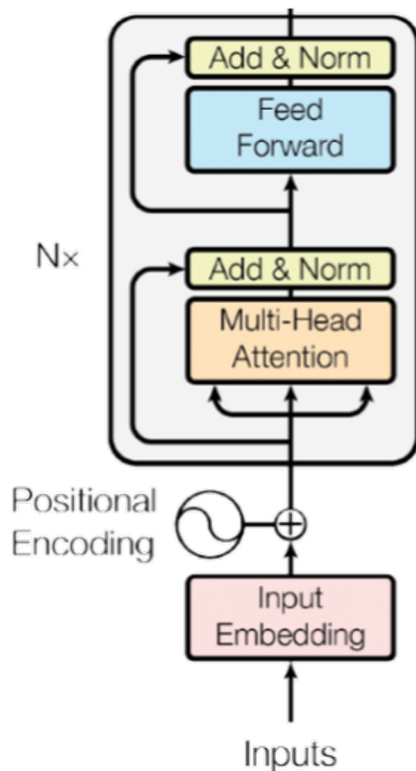


Final input representation = Token + Segment + Position embeddings



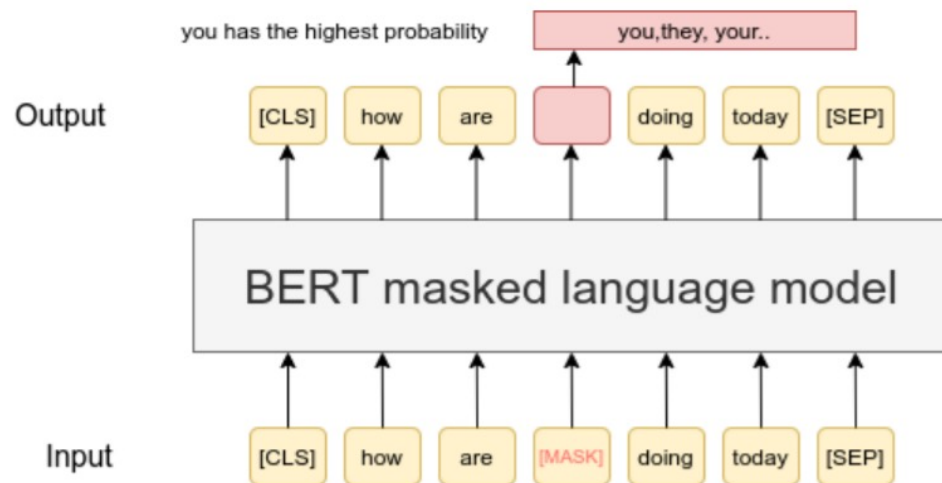
Separate two segments

BERT pre-training: Putting together



- **BERT-base**: 12 layers, 768 hidden size, 12 attention heads, 100M parameters
← Same as OpenAI GPT
- **BERT-large**: 24 layers, 1024 hidden size, 16 attention heads, 340M parameters
- **Training corpus**: Wikipedia (2.5B) + BooksCorpus (0.8B)
OpenAI GPT was trained on BooksCorpus only!
 ↓
- **Max sequence size**: 512 word pieces
- **Trained for 1M steps**, batch size 128k

BERT pre-training: Putting together



- MLM and NSP are **trained together**
- The [CLS] representation is trained for NSP
- Token-level representations are trained via MLM

Sentence-level Tasks

- Sentence pair classification tasks:

MNLI

Premise: A soccer game with multiple males playing.

Hypothesis: Some men are playing a sport. {entailment, contradiction, neutral}

QQP

Q1: Where can I learn to invest in stocks?

Q2: How can I learn more about stocks? {duplicate, not duplicate}

- Single sentence classification tasks:

SST2

rich veins of funny stuff in this movie

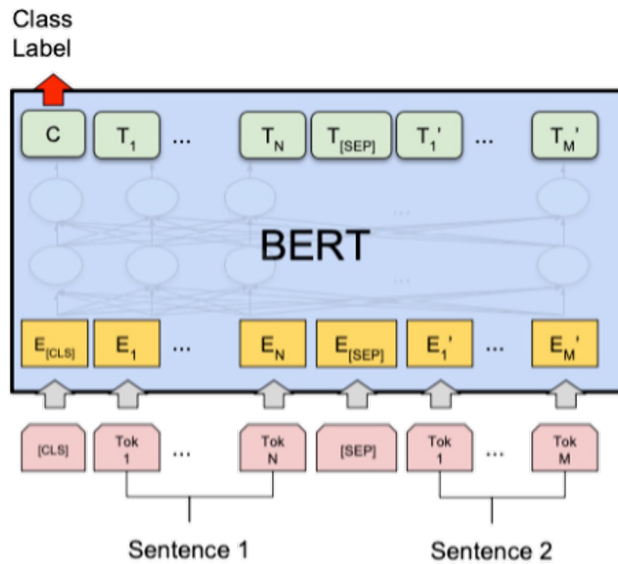
{positive, negative}

GLUE

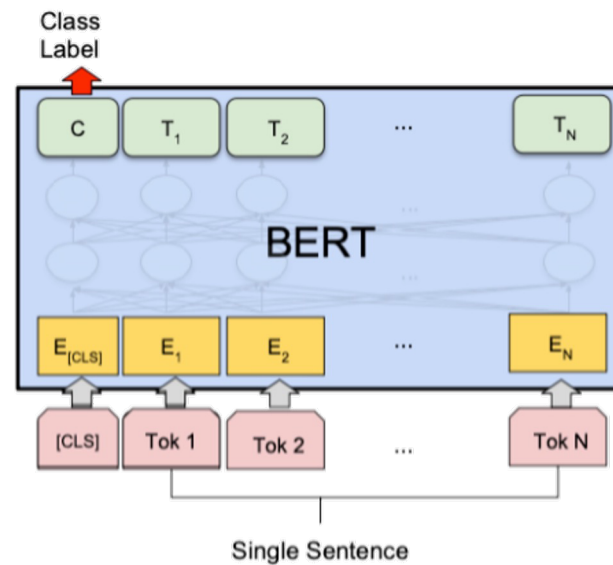
(Wang et al., 2018): 6 sentence pair and 2 single-sentence tasks

Fine-tuning BERT: Sentence-level tasks

“Pretrain once, finetune many times.”

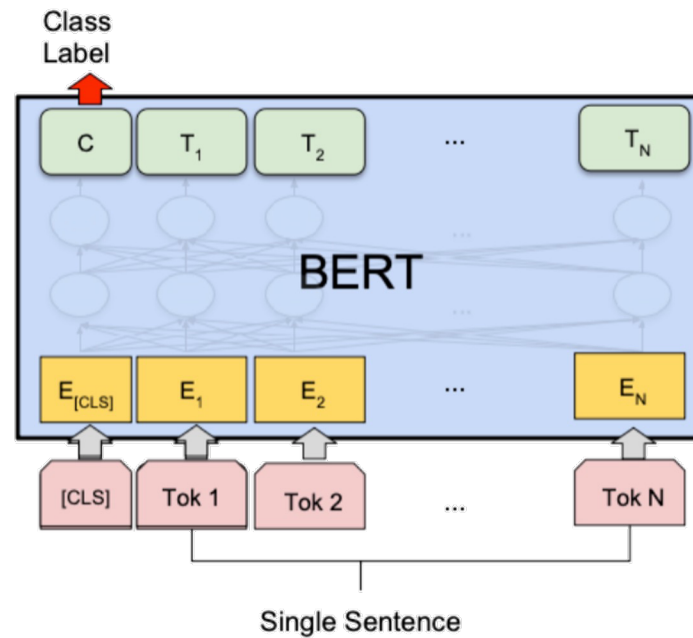
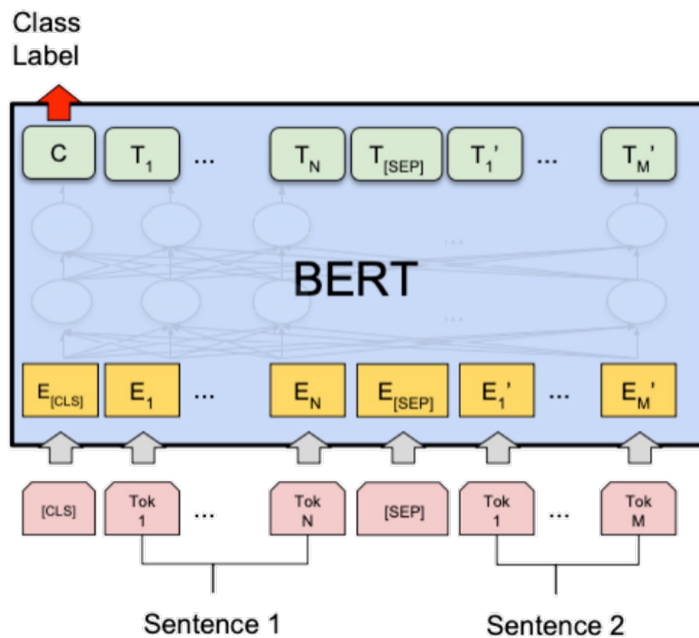


(a) Sentence Pair Classification Tasks:
MNLI, QQP, QNLI, STS-B, MRPC,
RTE, SWAG



(b) Single Sentence Classification Tasks:
SST-2, CoLA

Fine-tuning BERT



- For sentence-pair tasks, use [SEP] and segment embeddings to distinguish inputs
- Add a task-specific linear classifier on top of the [CLS] representation.

Token-level Tasks

- Extractive question answering e.g., SQuAD ([Rajpurkar et al., 2016](#))

SQuAD

Question: The New York Giants and the New York Jets play at which stadium in NYC ?

Context: The city is represented in the National Football League by the New York Giants and the New York Jets , although both teams play their home games at MetLife Stadium in nearby East Rutherford , New Jersey , which hosted Super Bowl XLVIII in 2014 .

(Training example 29,883)

MetLife Stadium

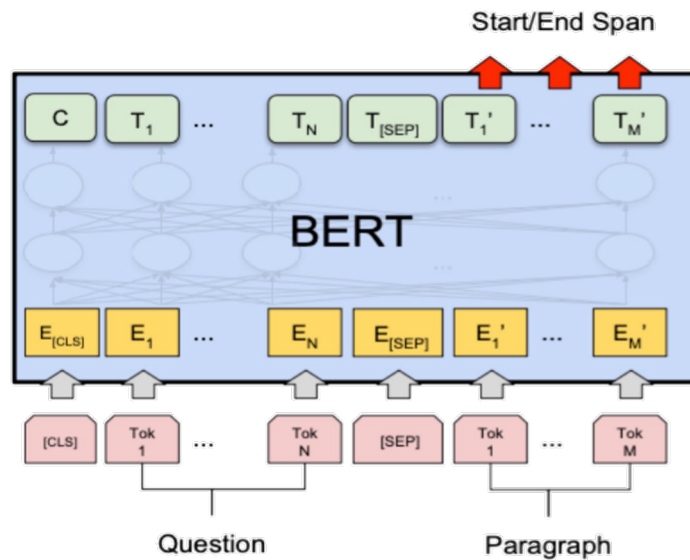
- Name entity recognition ([Tjong kim Sang and De Meulder, 2003](#))

CoNLL 2003 NER

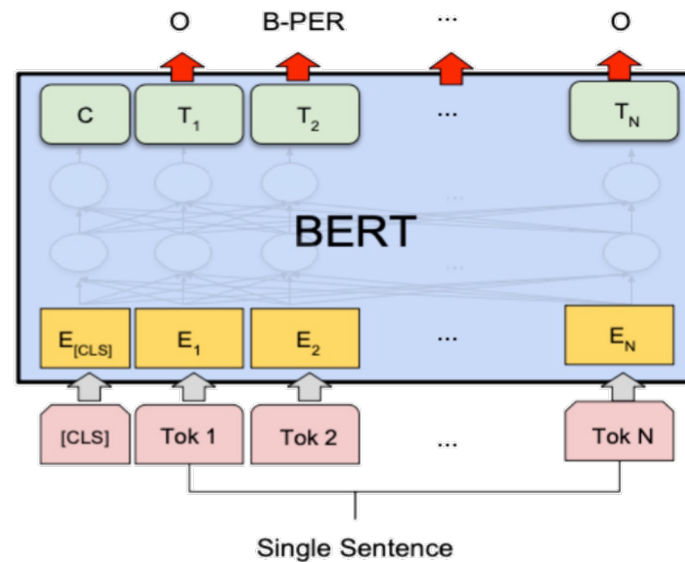
John	Smith	lives	in	New	York
B-PER	I-PER	O	O	B-LOC	I-LOC

Fine-tuning BERT: Token-level Tasks

“Pretrain once, finetune many times.”

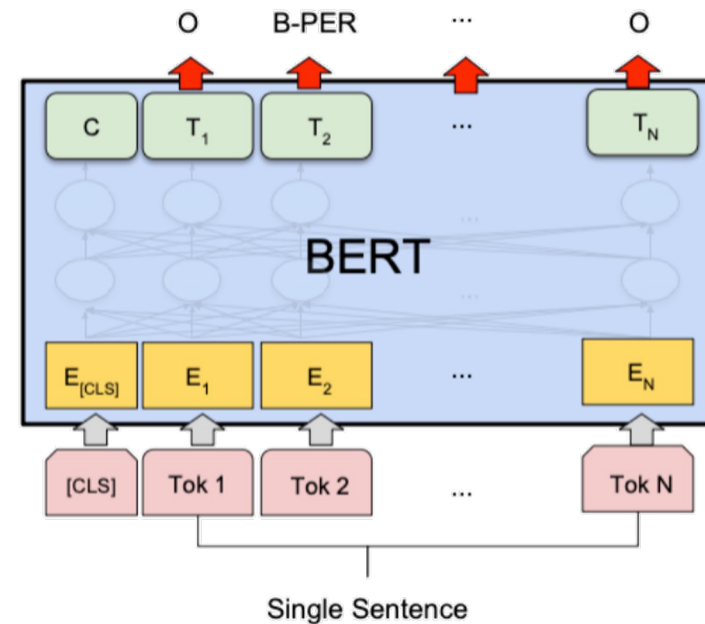
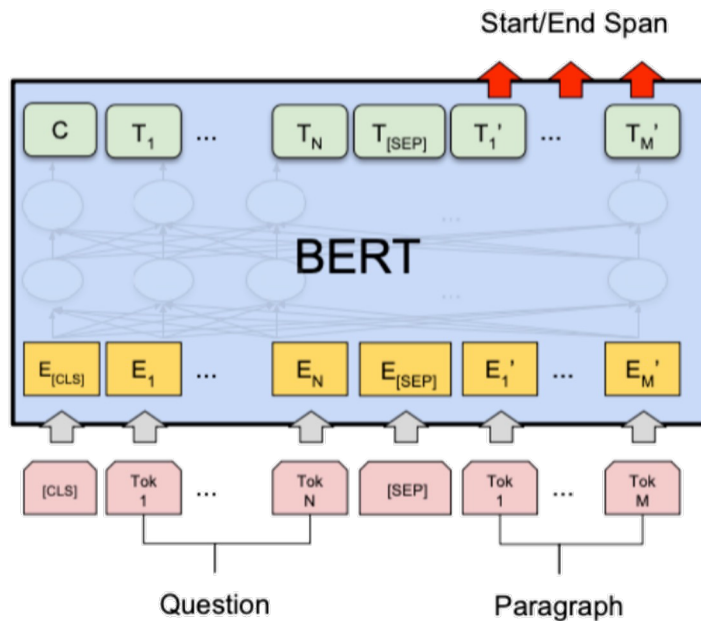


(c) Question Answering Tasks:
SQuAD v1.1



(d) Single Sentence Tagging Tasks:
CoNLL-2003 NER

Fine-tuning BERT



- For token-level prediction tasks, add a task-specific linear classifier on top of each token's hidden representation.

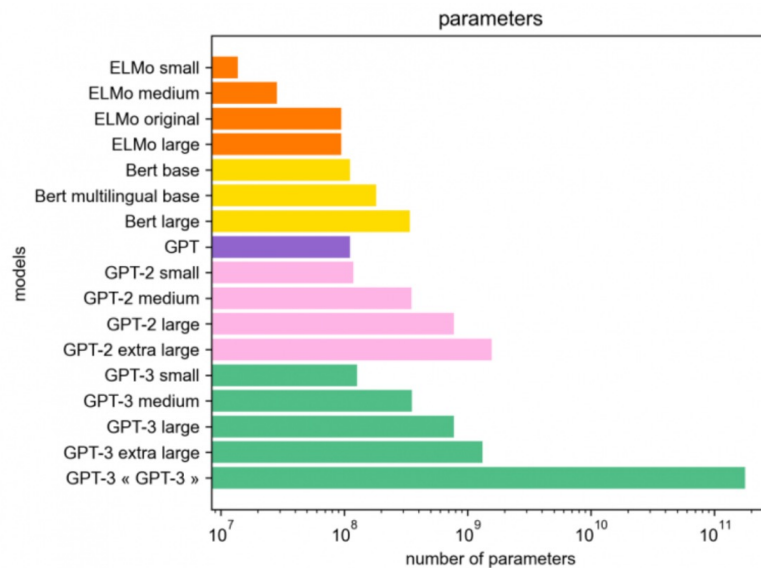
Ablation study: Model Sizes

Hyperparams				Dev Set Accuracy		
#L	#H	#A	LM (ppl)	MNLI-m	MRPC	SST-2
3	768	12	5.84	77.9	79.8	88.4
6	768	3	5.24	80.6	82.2	90.7
6	768	12	4.68	81.9	84.8	91.3
12	768	12	3.99	84.4	86.7	92.9
12	1024	16	3.54	85.7	86.9	93.3
24	1024	16	3.23	86.6	87.8	93.7

The bigger, the better!

?

How large are Large“ LMs?



More recent models: PaLM (540B), OPT (175B), BLOOM (176B)...

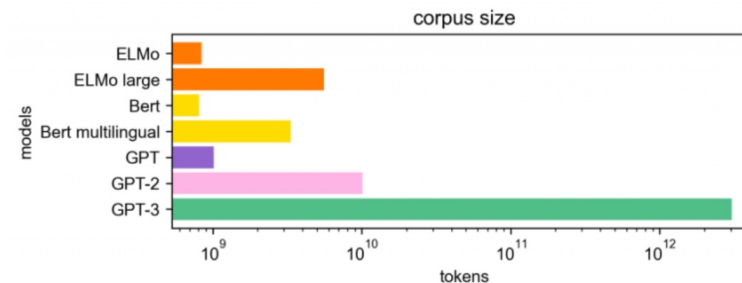


Image source: <https://hellofuture.orange.com/en/the-gpt-3-language-model-revolution-or-evolution/>

Summary

- BERT is a deep bidirectional Transformer encoder pre-trained using masked language modelling.
- Bidirectional self-attention allows each token to attend to both its left and right context.
- BERT popularised the large-scale pretrain–then-fine-tune paradigm in NLP.
- Achieved state-of-the-art performance (2018) across diverse NLU tasks.